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Viewpoint

Generative Artificial Intelligence: A New Engine for Advancing Environmental Science and Engineering

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Artificial intelligence (AI) technology has started to be integrated into our daily lives and work. Traditional discriminative AI focuses on learning decision boundaries between different categories of data and has already been widely applied in environmental science and engineering, such as water quality prediction and pollution detection.[1] Recently, the new frontier is generative AI, which generates new data instances based on learned data distributions. For instance, we can use large language models (LLMs) like ChatGPT to write, translate, and code, or we can use text-to-image models like DALL-E to create artistic works without prior drawing skills.[2]

As it rapidly progresses, generative AI is bringing new possibilities to scientific research for many fields.[3,4] In the face of global sustainability challenges such as climate change, access to clean water, and biodiversity loss, generative AI may offer innovative solutions to these interconnected and large-scale sustainability issues.[5]

This Viewpoint will discuss the current and potential applications and existing obstacles of generative AI in environmental science and engineering (Figure 1), aiming to inspire research in environmental generative AI and advance the discipline.

1. TOOL TO ENHANCE THE PERFORMANCE OF TRADITIONAL DISCRIMINATIVE AI

Data for environmental quality monitoring are difficult to acquire due to harsh monitoring environments and the high costs of sensing and labeling. The scarcity of data poses significant challenges to the construction of discriminative AI models for tasks such as event detection, especially for deep learning models that require large data sets.[6] Generative AI, such as generative adversarial networks (GANs), can enhance

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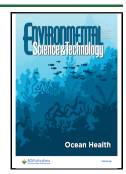

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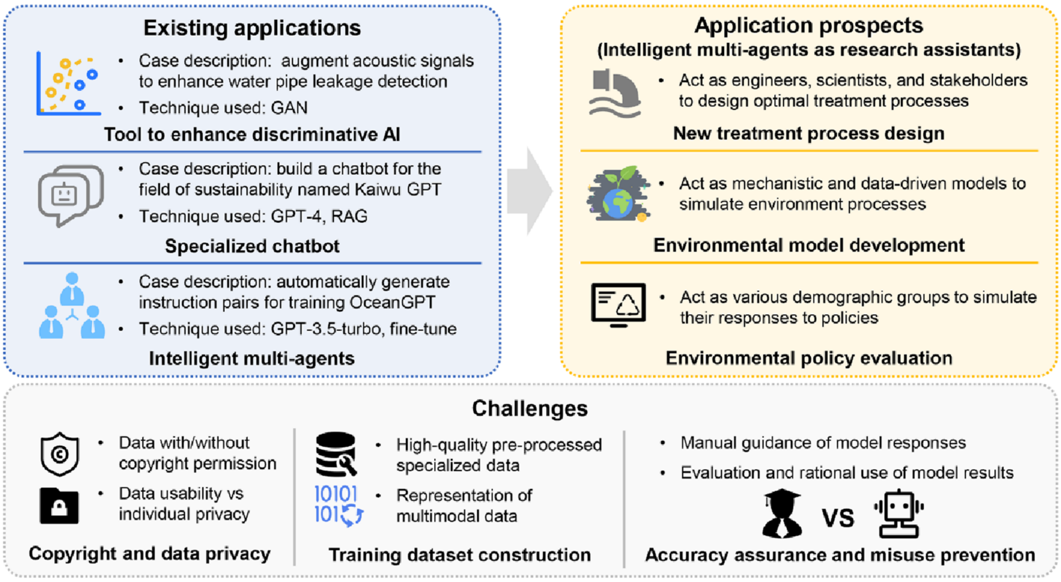

Figure 1. Current and potential applications and existing obstacles of generative AI in environmental science and engineering.

the performance of these data-hungry models by generating additional training data (e.g., time series and images) based on learned data distributions. For example, Liu et al.[7] used adversarial training between the generator and discriminator in a GAN for augmenting acoustic signals to train a water pipeline leakage detection model, which can save water resources and prevent water contamination through broken pipes. The generator is responsible for creating new samples, while the discriminator evaluates their authenticity. The generator continuously improves until its generated samples are indistinguishable from real data, thereby achieving data augmentation. The obtained results indicated that the augmented data improved the leakage detection performance. This represents a primary use of generative AI but still does not fully exploit the potential of generative AI.

2. LLM-BASED ENVIRONMENTAL CHATBOTS

Inspired by the interactive capabilities of LLMs like ChatGPT, we can build an environmental-specific LLM (i.e., a chatbot) using discipline-specific knowledge. This specialized LLM can serve as a knowledge question-and-answer repository for research, teaching, and training, helping environmental researchers and practitioners quickly acquire professional knowledge. Additionally, leveraging the in-context learning ability of LLMs, the chatbot can rapidly develop new capabilities, e.g., incorporating a few labeled Raman spectra indicating the presence or absence of organics in prompts to enable it to identify organic pollution. Furthermore, the chatbot can be embedded with tools through API calling, enabling users to call existing discriminative AI models or specialized tools (e.g., hydraulic models) via conversational interfaces rather than through coding. This requires human–model interaction, and the specificity and clarity of the prompts determine the quality of the model responses.[2,3]

Kaiwu GPT, which can give professional responses, analyze data, and code, is a typical example of environmental chatbots. It is developed on the basis of the TianGong AI open-source project initiated by Professor Ming Xu from Tsinghua University, focusing on the field of sustainability.[8] Built using the retrieval-augmented generation (RAG) framework, this chatbot leverages a LLM (GPT-4 in this case) to first retrieve the most relevant information from external data sources (such as papers and books in the field of sustainability) in response to a given prompt. On the basis of this retrieved information, the LLM then generates a response.[9] This allows the chatbot to enhance the accuracy and relevance of the generated content by utilizing external knowledge. Compared to fine-tuning a fundamental LLM directly on external data (which is often limited in size relative to the model's vast parameters), RAG has a significant advantage. It can generate more accurate answers in few-shot or even zero-shot scenarios by introducing relevant external documents. Moreover, RAG enables dynamic updates of the chatbot by updating external data sources without altering the fundamental LLM.

3. LLM-DRIVEN ENVIRONMENTAL INTELLIGENT AGENTS

The field of intelligent agents powered by LLMs is rapidly developing.[10] Specifically, agents are created by LLMs to be able to autonomously make decisions on the basis of the knowledge of LLMs and take actions (e.g., writing codes to call other applications through API) on their own. LLMs can process multimodal data (e.g., text, sound, and images), allowing the agent to perceive external environments or human instructions. After processing information, making decisions, reasoning, and planning, the agent can provide results and feedback in text form or directly call embedded tools. The key difference between this and a chatbot is that, given a task or environmental stimulus, the agent can autonomously plan and execute tasks. Beyond completing individual tasks autonomously, an LLM-driven agent can interact with other agents, demonstrating a certain level of social capability. This has led to more interest in multi-agent technology, which simulates interactions among multiple agents to foster knowledge creation and discovery, such as in software development, scientific discovery, and policy making.[11]

To the best of our knowledge, applications of intelligent agents for direct automated decision making in the environmental field have not yet been reported. However, some researchers have used the multi-agent collaboration to build

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Environ. Sci. Technol. 2024, 58, 17524−17528

environmental chatbots. When constructing OceanGPT, a model focused on issues like marine resource development and ecological protection, Bi et al.[12] did not use the RAG framework. Instead, they chose to build the model from scratch using data specific to the marine domain.

The creation of OceanGPT primarily involved two stages: pretraining and instruction fine-tuning. In the pretraining stage, self-supervised learning techniques were used to enable a LLM (Llama-2 in this case) to generate relevant information for the marine domain. The instruction fine-tuning stage then involved supervised training using paired question-and-answer instructions to bridge the gap between the LLM's text generation capabilities and the goal of answering domain-specific questions. To address the challenge of insufficient instructions, the study employed multi-agent technology (with agents powered by GPT-3.5-turbo) to automatically generate new instructions (more than 150 000 entries) based on a small, manually curated data set of instructions (10 000 entries created by experts over four days). Each agent was fine-tuned on marine domain data, with some agents tasked with enriching and refining existing instructions, others responsible for extracting new instructions directly from the training data set, and still others using a debate mechanism to control the quality of newly generated instructions. Through manual evaluation by experts, it was determined that the multi-agent collaboration resulted in instructions that were both professional and diverse. This automated instruction generation approach significantly reduced the amount of human labor and greatly improved the efficiency of specialized LLM development.

4. PROSPECTS FOR APPLICATIONS OF GENERATIVE AI IN ENVIRONMENTAL SCIENCE AND ENGINEERING

Currently, the application of generative AI in environmental science and engineering is mostly limited to generating training data and specialized chatbots. However, LLM-driven agents possess the capability for autonomous decision making, giving them the potential to become valuable assistants or partners for environmental researchers. When facing complex environmental problems, each LLM-driven agent can be considered as an individual or organization with a specific role, such as a scientist or government entity. Through interactions among these agents, we can mimic how various stakeholders perceive, interact, and make decisions and explore impacts of policies or adoption of new technologies, making it easier to accomplish various complex tasks.

4.1. Design New Treatment Processes. Against the backdrop of global climate change and severe pollution, it is important to design new processes for wastewater, exhaust gas, or solid waste treatment to reduce pollution and greenhouse gas emissions. LLM-driven multi-agent technology can simulate interactions among engineers who understand the practical problems in process operation, environmental scientists who grasp the principles of pollutant migration and transformation, and stakeholders who actually use the technology. Through iterative interactions between the human and LLM agents, they can develop potential solutions that are both theoretically sound and practically feasible.

4.2. Develop Environmental Models. Simulating pollutant diffusion is challenging due to the heterogeneity of environmental media and the complexity of diffusion processes. The construction of mechanistic models is complex and often hampered by unrealistic assumptions or simplifications about environmental systems and high parameter uncertainty, leading to poor simulation results. While data-driven models are easier to construct, they perform poorly with insufficient data and lack interpretability.[13] As such, LLM-driven multi-agent technology can act as a "control center" to operate various models, mechanistic or data-driven, autonomously to utilize the best that these models can offer.

4.3. Evaluate Environmental Policies. To make informed and responsible decisions, ensure policy effectiveness, and avoid unintended consequences, it is essential to evaluate policies beforehand. LLM-driven multi-agent technology is well-suited for this task.[11] For example, when formulating household energy-saving policies, agents can be trained with behavioral information from various demographic groups to accurately simulate interactions among different households and their responses to the policies. This provides valuable insights for policymakers on policy formulation and its potential impacts.

5. CHALLENGES

Generative AI model development is advancing rapidly, with numerous open-source models and tools like Llama[14] readily available. Building the models themselves is becoming a smaller hurdle, especially when collaborating with experts in computer science, which can significantly enhance efficiency. However, for researchers in the environmental field, major challenges lie in acquiring and creating specialized data sets before model construction and ensuring the accuracy of outputs during model development and usage.

5.1. Copyright and Data Privacy. Ensuring model accuracy often necessitates large amounts of data from various sources. Issues of copyright and data privacy directly impact the amount of usable data we can access. Data may be subject to copyright restrictions, raising the question of whether they can be used for training without the permission of the copyright holders. Additionally, large models remember the training samples and inadvertently disclose sensitive information. For instance, using personal electricity and water usage data to construct agents representing different demographic groups can pose privacy risks. To address copyright issues, we can prioritize the use of open-source data, as seen in the OceanGPT project,[12] but it is still essential to obtain copyright permissions wherever possible to increase the volume of training data and enhance model accuracy. With regard to data privacy concerns, we can attempt to balance data usability with individual privacy by anonymizing data (e.g., removing personal information during preprocessing) or introducing controlled noise.[15] Additionally, we can employ output monitoring tools, such as GPT-4's rule-based reward model, to monitor and filter sensitive information, thereby preventing privacy breaches.[16]

5.2. Training Data Set Construction. Environmental data are complex and diverse, including manual records, electronic prints, engineering drawings, time series, and images. With the extensive amount of raw data, the challenge lies in how to preprocess them and ultimately produce high-quality training data, as this process is often time-consuming and demands significant human effort and financial resources. Data preprocessing may involve removing irrelevant information (e.g., page numbers, footnotes, and URLs), deduplication, and ensuring information consistency. To ensure data quality, specialized processes or tools may need to be designed, such as

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Environ. Sci. Technol. 2024, 58, 17524−17528

involving environmental experts (or trained staff) to evaluate the accuracy of randomly sampled data.[12] Finally, for multimodal inputs, it is necessary to align the representations of visual and other modalities (i.e., vectors generated by encoders) with the feature space of LLMs.[17] For example, this involves ensuring that the representation of a lake with a green, paint-like layer on its surface in an image is closely related to the representation of the term "algal bloom" in semantic space. This alignment ensures that the training data are in a format that the model can effectively process. It is worth noting that this alignment process requires the deep involvement of environmental experts to manually create high-quality, cross-modal training samples, such as an image−text pair showing and describing the algal bloom.

5.3. Accuracy Assurance and Misuse Prevention. Generative AI technology can sometimes produce incorrect results, and if used indiscriminately, it can lead to the publication and dissemination of erroneous research findings.[2,5] To ensure accurate and reliable results, human involvement is crucial throughout the development and use of these models. In model development, a standardized training data construction process should be established to ensure its quality. Additionally, expert-guided models based on reinforcement learning can be developed, in which experts in the environmental field score an LLM's responses to guide the LLM toward more accurate results. Techniques such as RAG and instruction fine-tuning, as mentioned above, can also be employed to improve the accuracy of model outputs. In model usage, a feedback mechanism should be established so that humans can continuously provide evaluations of the accuracy of results, which will also aid in the development of expert-guided models. Furthermore, for high-risk or sensitive areas (e.g., environmental issues closely related to public welfare), results should be subject to mandatory expert review.[2] In summary, generative AI should be viewed as a tool to enhance efficiency but not as a replacement for human-centered research.

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Notes

The authors declare no competing financial interest.

Biography

Shuming Liu is a Professor and the Dean of the School of Environment at Tsinghua University. His research focus on AI-driven urban water infrastructure management, sustainable and resilient urban water systems, and the water−energy nexus. Prof. Liu is the editor-in-chief of *AQUA - Water Infrastructure, Ecosystems and Society* and an associate editor of journals like *Urban Water Journal* and *Smart Water*.

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